

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Visualization Design Activity

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO5: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication.

Assessment Tool Weightage: 25%

Total Marks: 20

Code of Conduct

I **Naresh M(192424381)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Naresh M

Visualization Design Activity Rubric

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Data Preparation	20	Thoroughly imports, explores, and prepares the industry dataset; correctly handles data types, missing values, and relevant variables	Imports and explores the data correctly with minor gaps in preparation	Performs basic data exploration and preparation but misses some important aspects	Unable to correctly import, explore, or prepare the dataset
Implementation	20	Efficiently applies Pandas and Python concepts for filtering, grouping, aggregation, transformation, and analysis	Applies appropriate Pandas operations with minor coding or analytical errors	Uses basic Pandas operations but several required analytical tasks are incomplete	Demonstrates limited understanding of Pandas
Visualization Techniques	25	Selects highly appropriate visualizations plots, and plot components	Uses appropriate visualizations for most requirements with minor errors or formatting issues	Uses basic visualizations but some are inappropriate, incomplete, or poorly formatted	Uses inappropriate or insufficient visualizations
Pattern Identification	20	Identifies meaningful industry-specific trends, relationships,	Identifies relevant patterns and provides logical interpretations	Identifies some basic patterns but provides limited or weak interpretation	Fails to identify meaningful patterns or provides

		distributions, and patterns	with adequate evidence		incorrect/unsupported interpretations
Recommendations	15	Converts analytical findings into practical, feasible industry recommendations	Provides relevant recommendations and presents findings clearly with minor gaps	Provides basic recommendations with limited connection to analytical findings	Recommendations are missing, impractical, or unrelated to the analysis; poor presentation

Total Marks Awarded:

Signature of the Course Faculty

Questions:

5. Smart City Traffic Visualization

A smart-city system collects traffic information containing Date_Time, Location, Vehicle_Type, Vehicle_Count, Average_Speed, Traffic_Density, and Weather. Develop a Python visualization solution to help city traffic authorities understand traffic patterns. Analyze the evolution of vehicle counts and traffic density over time, compare traffic volume across locations and vehicle types, investigate the relationship between traffic density and average speed, and visualize the distribution of vehicle counts. Design a set of clear and informative visualizations using Pandas and Matplotlib, applying appropriate titles, labels, legends, scales, figure sizes, and layouts. Interpret the resulting patterns to identify peak traffic periods and locations and suggest practical measures for improving traffic management.

CODE:

```
import pandas as pd

import matplotlib.pyplot as plt

df = pd.read_csv("smart_city_traffic.csv")

df["Date_Time"] = pd.to_datetime(df["Date_Time"])

print("First 5 Records:")

print(df.head())

print("\nDataset Information:")

print(df.info())

print("\nStatistical Summary:")
```

```

print(df.describe())
print("\nMissing Values:")
print(df.isnull().sum())
print("\nVehicle Type Counts:")
print(df["Vehicle_Type"].value_counts())
print("\nLocation Counts:")
print(df["Location"].value_counts())
time_data = df.groupby("Date_Time")[["Vehicle_Count", "Average_Speed"]].mean()
plt.figure(figsize=(10, 5))
plt.plot(time_data.index, time_data["Vehicle_Count"], marker="o")
plt.title("Evolution of Vehicle Counts Over Time")
plt.xlabel("Date and Time")
plt.ylabel("Vehicle Count")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
density_data = df.groupby("Date_Time")["Vehicle_Count"].mean()
plt.figure(figsize=(10, 5))
plt.plot(
    density_data.index,
    density_data.values,
    marker="o",
    label="Vehicle Count"
)
plt.title("Traffic Volume Over Time")
plt.xlabel("Date and Time")
plt.ylabel("Average Vehicle Count")
plt.legend()
plt.xticks(rotation=45)

```

```
plt.tight_layout()
plt.show()
location_volume = df.groupby("Location")["Vehicle_Count"].sum()
plt.figure(figsize=(8, 5))
location_volume.plot(kind="bar")
plt.title("Traffic Volume Across Locations")
plt.xlabel("Location")
plt.ylabel("Total Vehicle Count")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
vehicle_volume = df.groupby("Vehicle_Type")["Vehicle_Count"].sum()
plt.figure(figsize=(8, 5))
vehicle_volume.plot(kind="bar")
plt.title("Traffic Volume by Vehicle Type")
plt.xlabel("Vehicle Type")
plt.ylabel("Total Vehicle Count")
plt.tight_layout()
plt.show()
plt.figure(figsize=(8, 5))
plt.scatter(
    df["Traffic_Density"].map(
        {"Low": 1, "Medium": 2, "High": 3}
    ),
    df["Average_Speed"]
)
plt.title("Traffic Density vs Average Speed")
plt.xlabel("Traffic Density (Low=1, Medium=2, High=3)")
plt.ylabel("Average Speed")
```

```
plt.tight_layout()
plt.show()
plt.figure(figsize=(8, 5))
plt.hist(df["Vehicle_Count"], bins=8, edgecolor="black")
plt.title("Distribution of Vehicle Counts")
plt.xlabel("Vehicle Count")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

weather_counts = df["Weather"].value_counts()
plt.figure(figsize=(7, 5))
weather_counts.plot(kind="bar")
plt.title("Traffic Observations by Weather Condition")
plt.xlabel("Weather")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

density_counts = df["Traffic_Density"].value_counts()
plt.figure(figsize=(7, 5))
density_counts.plot(kind="bar")
plt.title("Traffic Density Frequency")
plt.xlabel("Traffic Density")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

print("\nTraffic Volume by Location:")
print(location_volume)

print("\nTraffic Volume by Vehicle Type:")
print(vehicle_volume)
```

```

print("\nAverage Speed by Traffic Density:")
print(df.groupby("Traffic_Density")["Average_Speed"].mean())

peak_row = df.loc[df["Vehicle_Count"].idxmax()]

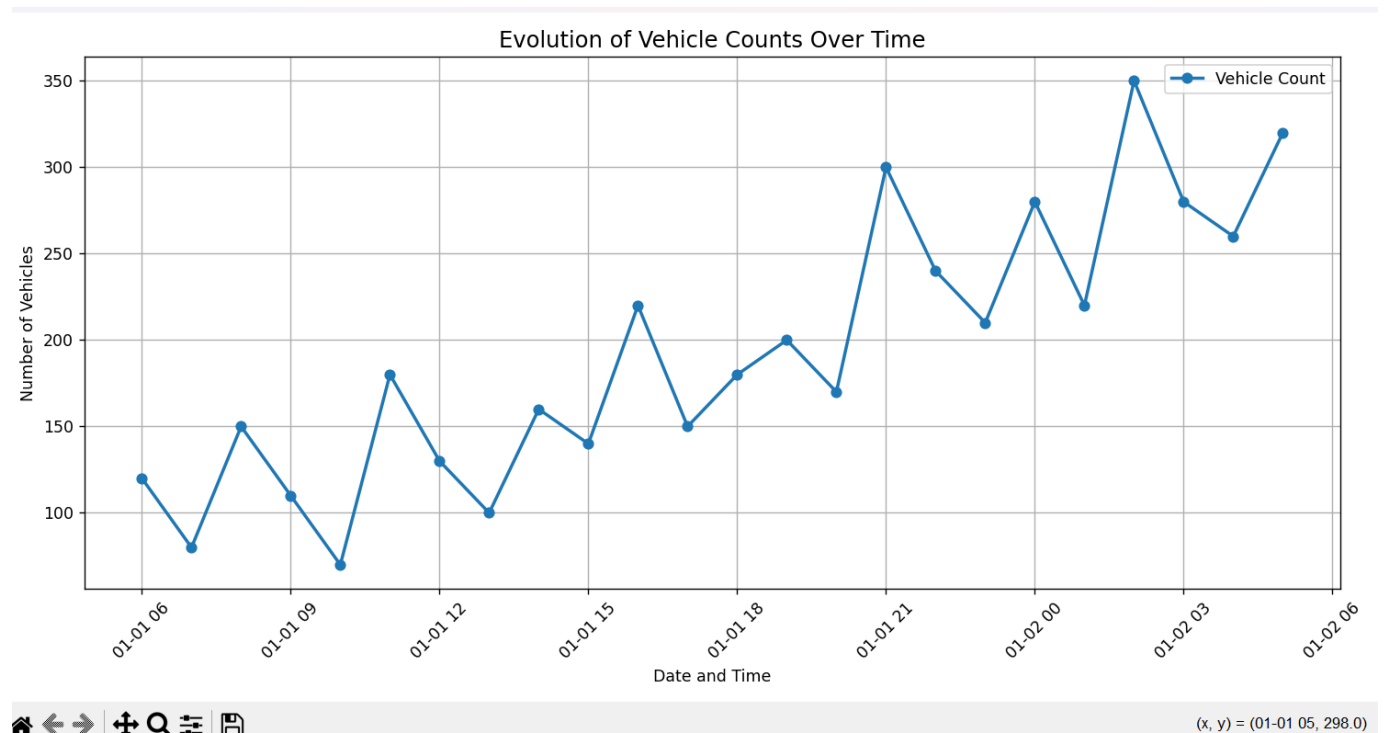
print("\nPeak Traffic Observation:")

print(peak_row)

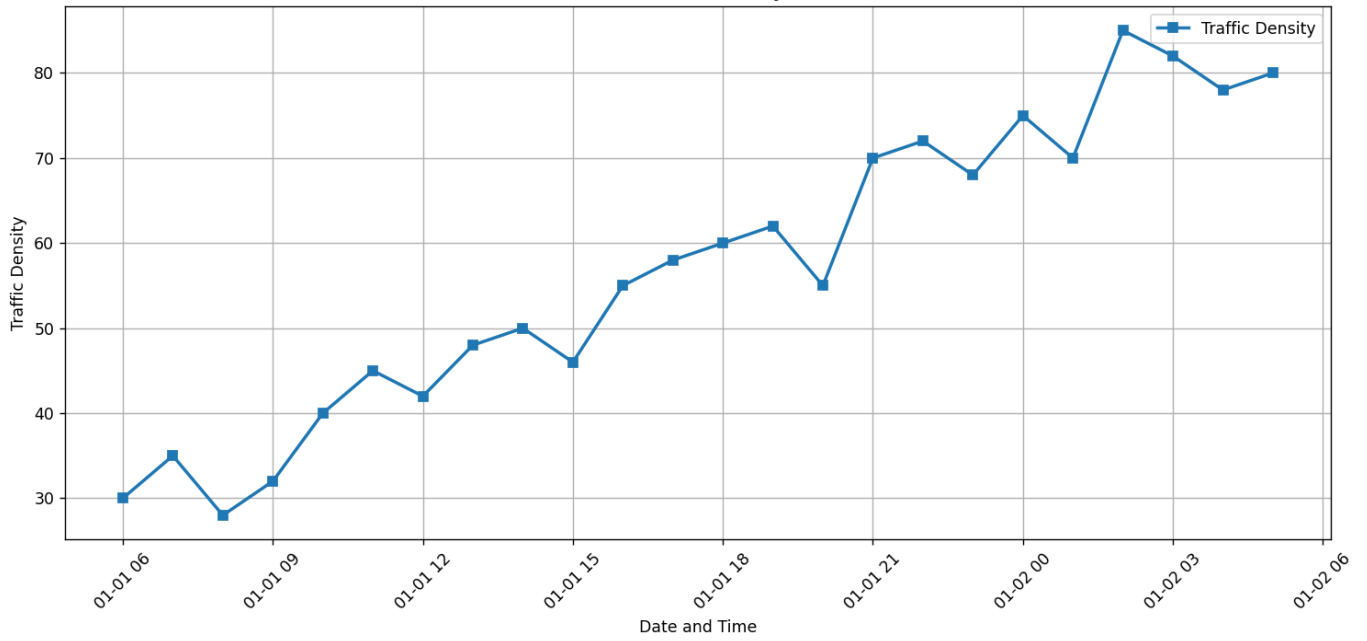
print("\nAverage Vehicle Count by Location:")
print(df.groupby("Location")["Vehicle_Count"].mean())

```

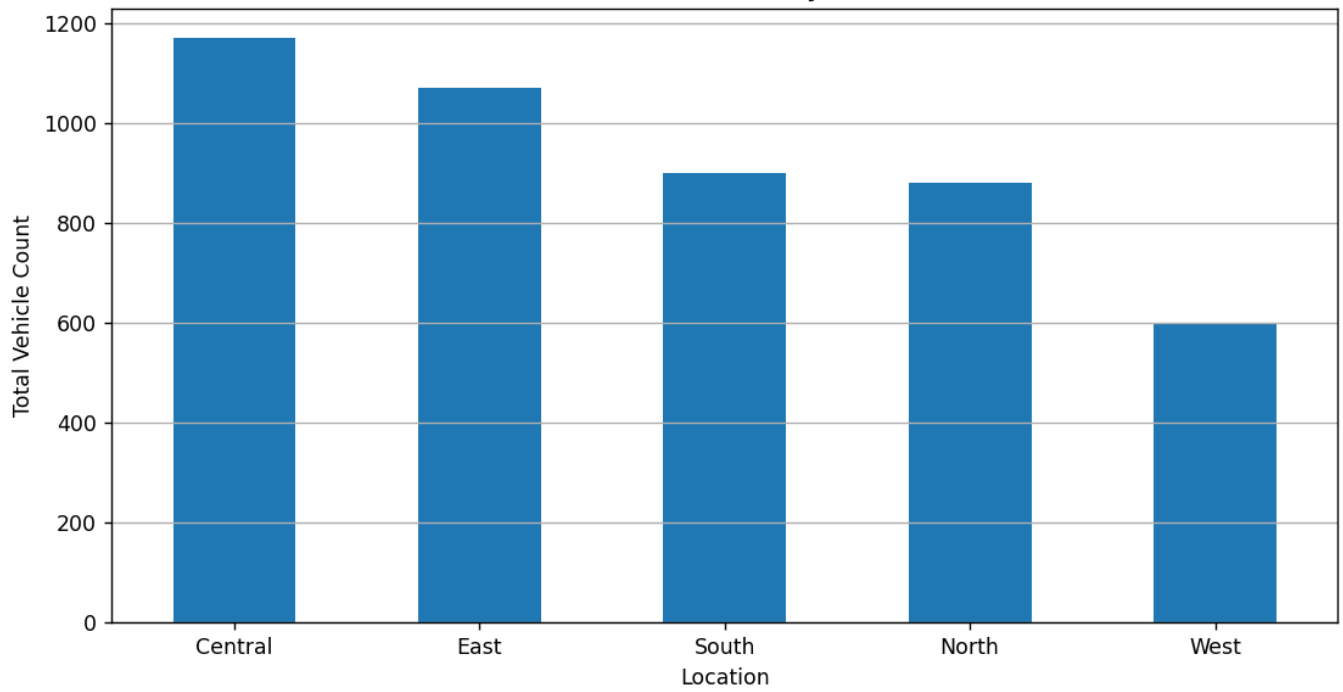
OUTPUT:



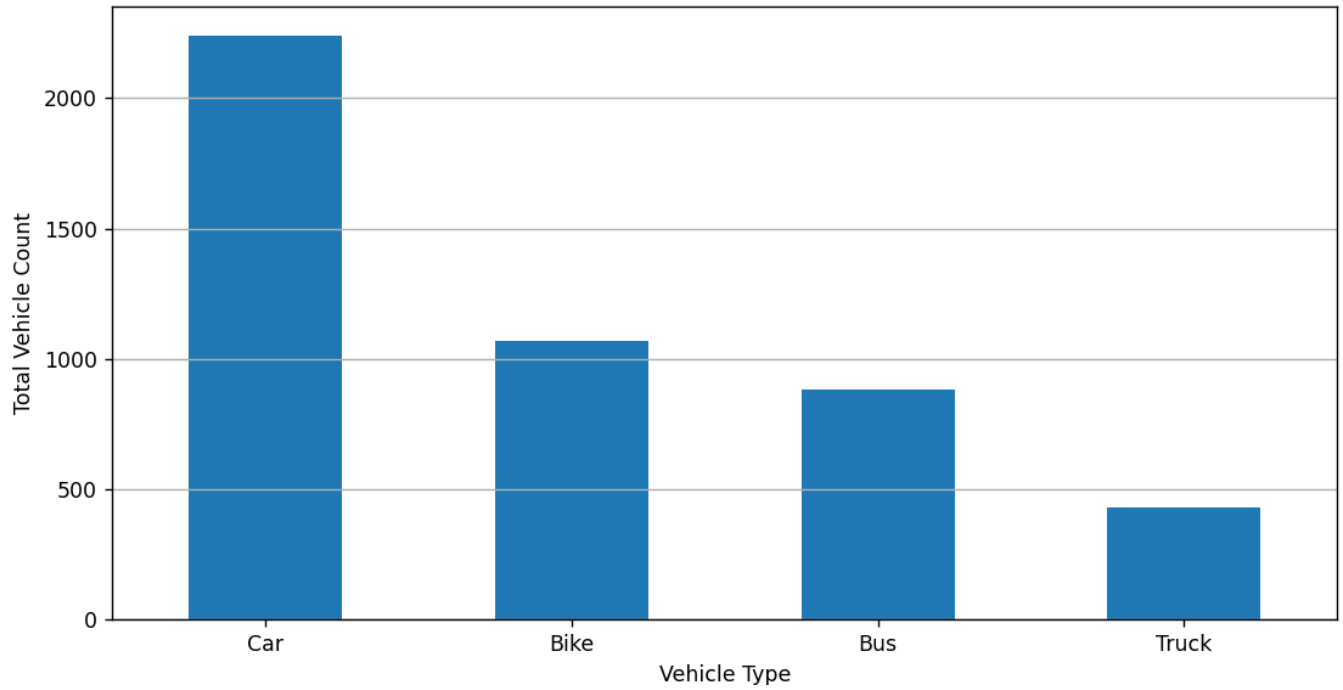
Evolution of Traffic Density Over Time



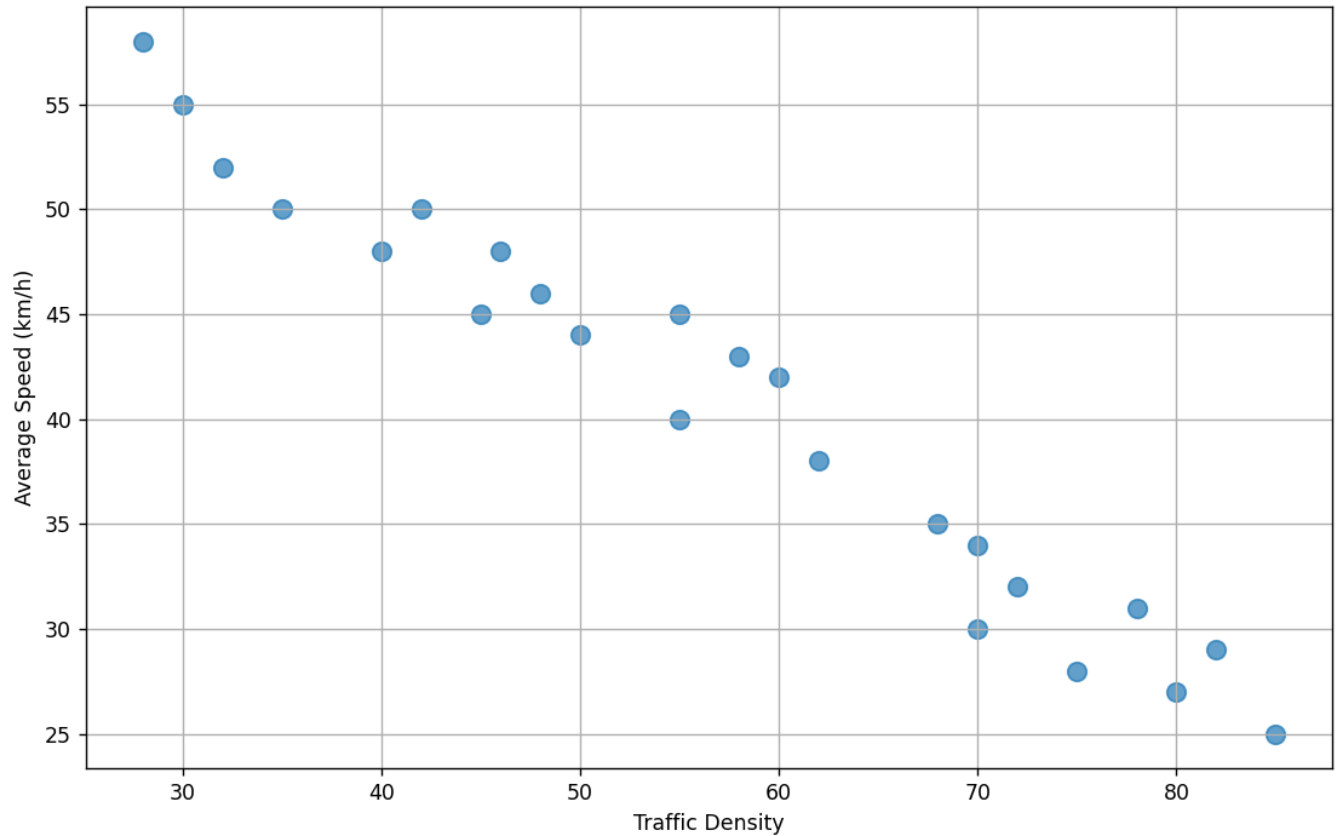
Total Traffic Volume by Location



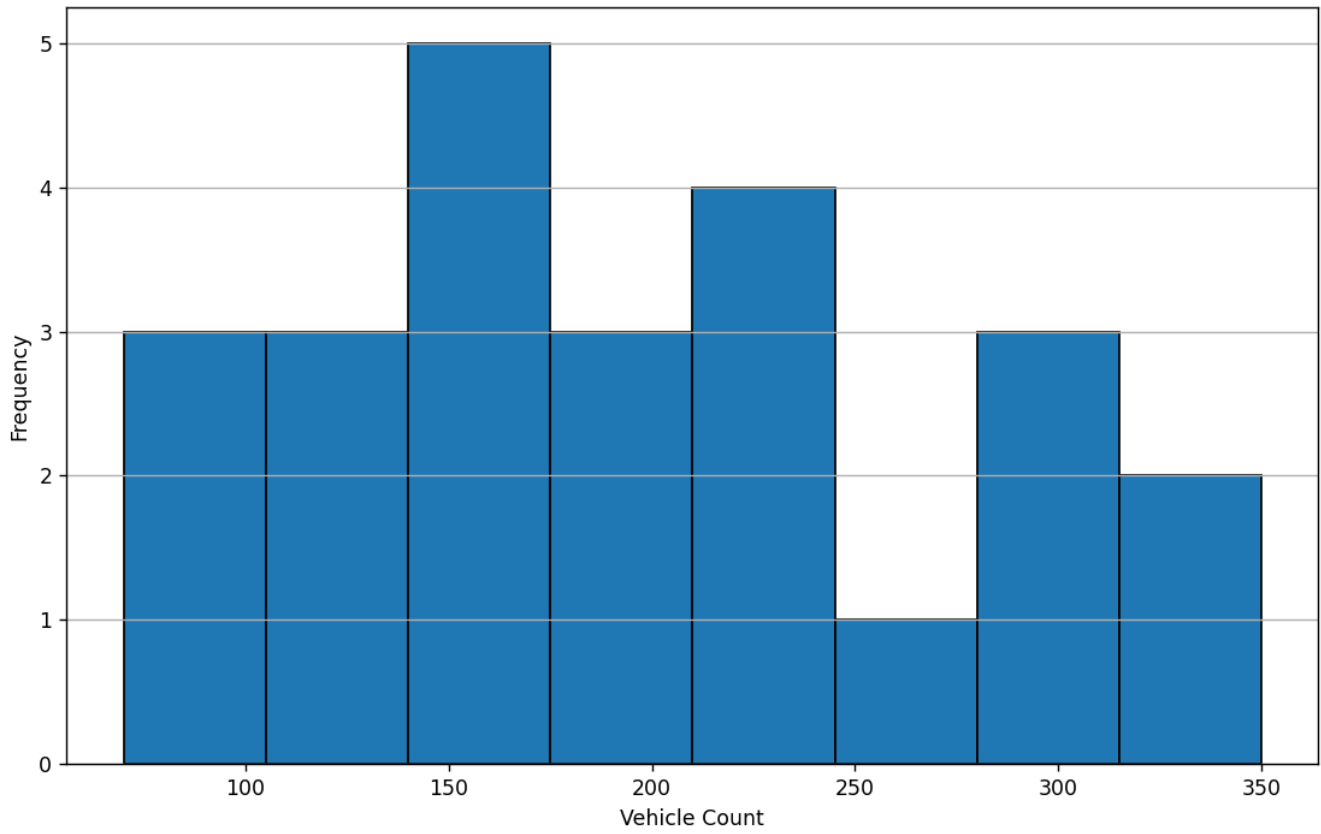
Traffic Volume by Vehicle Type



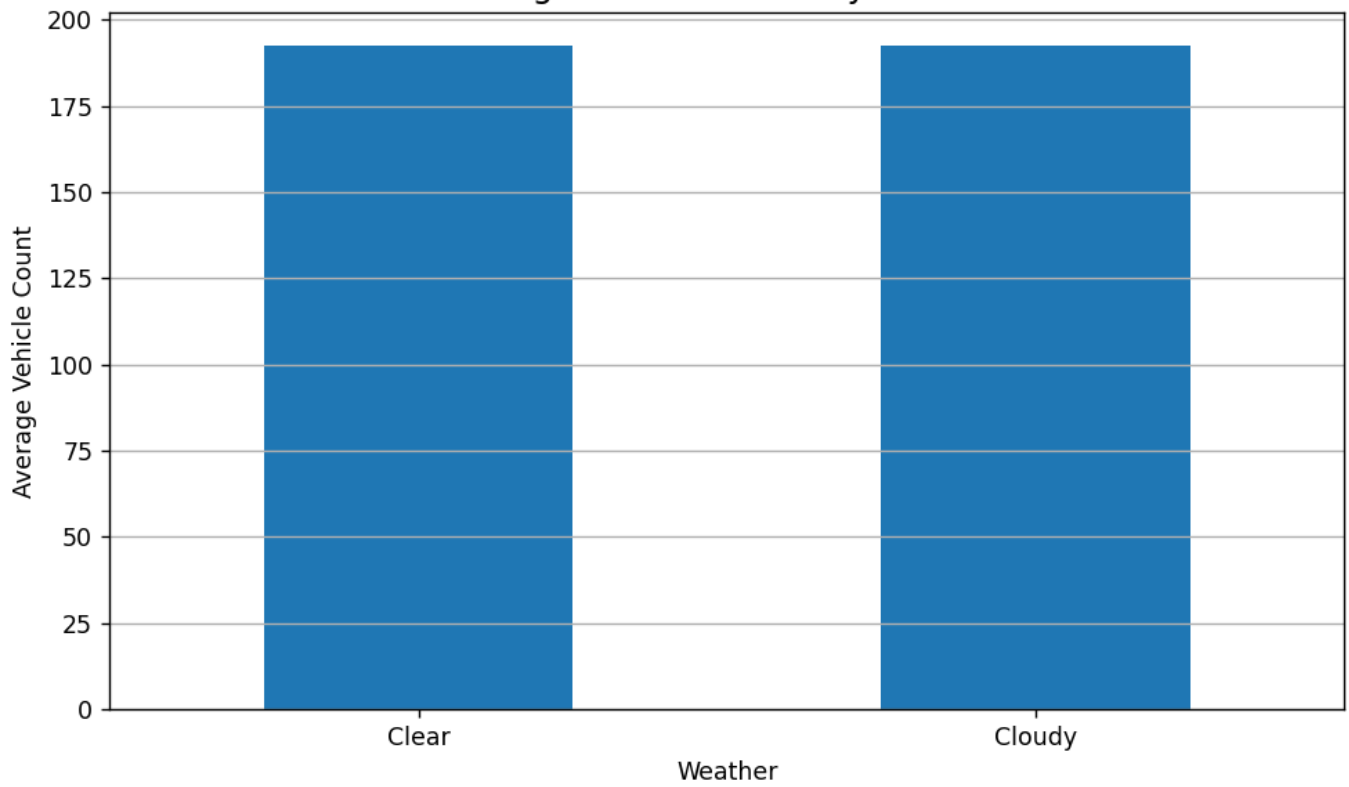
Relationship Between Traffic Density and Average Speed



Distribution of Vehicle Counts



Average Vehicle Count by Weather



First 5 Records:

	Date_Time	Location	Traffic_Density	Weather
0	2026-01-01 06:00:00	Central Junction	Low	Clear
1	2026-01-01 08:00:00	Central Junction	High	Clear
2	2026-01-01 10:00:00	Central Junction	Medium	Clear
3	2026-01-01 12:00:00	Central Junction	Medium	Cloudy
4	2026-01-01 14:00:00	Central Junction	Low	Clear

[5 rows x 7 columns]

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 32 entries, 0 to 31

Data columns (total 7 columns):

#	Column	Non-Null Count	Dtype
0	Date_Time	32 non-null	datetime64[us]
1	Location	32 non-null	str
2	Vehicle_Type	32 non-null	str
3	Vehicle_Count	32 non-null	int64
4	Average_Speed	32 non-null	int64
5	Traffic_Density	32 non-null	str
6	Weather	32 non-null	str

dtypes: datetime64[us](1), int64(2), str(4)

memory usage: 1.9 KB

None

Statistical Summary:

	Date_Time	Vehicle_Count	Average_Speed
count	32	32.000000	32.00000
mean	2026-01-03 01:22:30	278.437500	34.96875
min	2026-01-01 06:00:00	90.000000	22.00000
25%	2026-01-02 03:45:00	157.500000	29.50000
50%	2026-01-03 01:30:00	225.000000	34.50000
75%	2026-01-03 23:15:00	397.500000	40.00000
max	2026-01-04 21:00:00	560.000000	52.00000
std	NaN	143.372049	7.93719

Missing Values:

Date_Time	0
Location	0
Vehicle_Type	0
Vehicle_Count	0
Average_Speed	0
Traffic_Density	0
Weather	0

dtype: int64

Vehicle Type Counts:

Vehicle_Type	
Car	12
Bike	8
Bus	8
Truck	4

Name: count, dtype: int64

Location Counts:

Location	
Central Junction	8
Market Road	8
Airport Road	8
Tech Park	8

Name: count, dtype: int64

Traffic Volume by Location:

Location

Airport Road 2080

Central Junction 1970

Market Road 2330

Tech Park 2530

Name: Vehicle_Count, dtype: int64

Traffic Volume by Vehicle Type:

Vehicle_Type

Bike 2740

Bus 1510

Car 4240

Truck 420

Name: Vehicle_Count, dtype: int64

Average Speed by Traffic Density:

Traffic_Density

High 26.727273

Low 45.000000

Medium 35.769231

Name: Average_Speed, dtype: float64

Peak Traffic Observation:

Date_Time 2026-01-04 17:00:00

Location Tech Park

Vehicle_Type Car

Vehicle_Count 560

Average_Speed 22

Traffic_Density High

Weather Rain

Name: 29, dtype: object

Average Vehicle Count by Location:

Location

Airport Road 260.00

Central Junction 246.25

Market Road 291.25

Tech Park 316.25

Name: Vehicle_Count, dtype: float64